

Assignment 10: Array List (Vector)

***Build a Vector Class. Then, add the following Methods to it.**

- 1. Write a method that allows the user to reverse the order of elements of a given Vector.**

Test Data:

Original Vector: 1 2 3 4

Expected Output Vector: 4 3 2 1

Original Vector: apple banana cherry

Expected Output Vector: cherry banana apple

- 2. Remove Duplicates from a Vector.**

Write a method that takes a vector and returns a vector with the unique values only in the same order.

Test Data:

Original Vector: 6 1 8 1 9 2 3 6

Expected Output Vector: 8 9 2 3

Original Vector: black red red green yellow blue

Expected Output Vector: black green yellow blue

- 3. Insert a Number at a Specific Index**

Write a method that allows the user to insert an element at any specified position within a Vector.

Test Data:

Original Vector: 1 2 4 5 7

Insert an element with value 99 at position 2

Output: 1 2 99 4 5 7

4. Capitalize First Letter of Each Vector String

Write a method that allows the user to capitalize the first character of each element of a given string vector. Return the new vector.

Test Data:

Original Vector: red green black white Pink

Expected Output: : Red Green Black White Pink

5. Elements Smaller Than Adjacent Neighbors

Write a method that allows the user to return the elements in a vector that are strictly smaller than their adjacent left and right neighbors.

Test Data:

Original Vector: 7, 2, 5, 3, 1, 5, 6

Expected Output: : 2, 1

Original Vector: 1, 2, 5, 0, 3, 1, 7

Expected Output: : 0, 1

Bonus

Create a vector of pairs of any size where each pair holds a student's name and grade. Print the name of the student with the highest grade.

Input :

Sama 95

Menna 90

Maher 100

Ramzy 70

Output :

Maher